What Determines the Effectiveness of 3D Field of Junctions?

A Study of Volumetric Denoising Applications

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Abstract. 3D Field of Junctions (FoJ) is a training-free denoiser that fits each patch of a volume with planar wedges meeting at a common vertex and averages the overlapping fits. Its demonstrated applications are low-dose X-ray CT, cryo-electron tomography, and point-cloud denoising. We apply 3D FoJ to three domains outside that set, porous-media micro-CT, an MRI phantom, and binary porous media, and compare it with Gaussian, median, and total-variation (TV) filtering across noise levels to examine when the method helps and when it fails. At its default configuration, the method reduces to an independent wedge fit per patch followed by averaging, with no inter-patch coupling. From this mechanism follow three requirements for FoJ to help. First, the input must be unbiased: each region intensity is a mean of the voxels it covers, so a systematic offset in the input is reproduced rather than removed. Second, the structure must be piecewise-flat at or above the patch scale; features smaller than the patch cannot be represented by wedges, and the output collapses toward the mean. Third, the input noise must be high enough that the wedge model's bias is paid for by the noise it removes.

1 Introduction

Volumetric images acquired under dose and time limits are noisy, and denoising is a routine step in their analysis. Classical denoisers include local filters such as Gaussian smoothing and median filtering, variational methods such as total-variation (TV) minimization [1, 2], and patch-recurrence methods such as non-local means [3] and BM3D [4]; learned denoisers trained on paired data [5] dominate at moderate noise levels.

Field of Junctions (FoJ) [6] is a training-free structural prior that models each image patch as constant-intensity regions separated by boundaries meeting at a vertex, called a junction. 3D FoJ [7] extends the model to volumes: three planes meeting at a common vertex divide each cubic patch into constant-valued wedges, and overlapping patch fits are averaged into a denoised volume. The method targets the low-SNR regime and was demonstrated on low-dose X-ray CT, cryo-electron tomography, and point-cloud denoising.

We apply the released 3D FoJ implementation to three domains outside that set. Grayscale micro-CT of porous rock is a natural fit for the wedge model, since grains and pores form near-constant regions with sharp, piecewise-planar boundaries. An MRI phantom and two binary porous-media volumes are poor fits: the first has smooth tissue intensities, the second has structure smaller than the patch scale. Testing all three under controlled noise shows where the method transfers. Our implementation and experiment scripts are available in the accompanying repository.[1]

*Equal contribution; order determined by coin flip.

[1] https://github.com/samudraneel05/foj3d-factor-study

2 Methodology

Let $y \in \mathbb{R}^{N^3}$ be a noisy volume and let $\{\Omega_p\}_p$ be cubic patches of side R on a grid with stride s. Inside each patch p, three planes meeting at a shared vertex (parameterized by θ_p, consisting of three plane orientations and a vertex position) divide the patch into M wedge-shaped regions with soft indicator functions $u_{p,k}(\cdot;\theta_p)$ (Fig. 1). Given θ_p, each region takes a constant intensity given by the indicator-weighted mean of the voxels it covers,

$$c_{p,k} = \frac{\sum_{x\in\Omega_p} u_{p,k}(x)\, y(x)}{\sum_{x\in\Omega_p} u_{p,k}(x)}, \tag{1}$$

and the local patch rendering is $\hat{y}_p(x) = \sum_k u_{p,k}(x)\, c_{p,k}$.

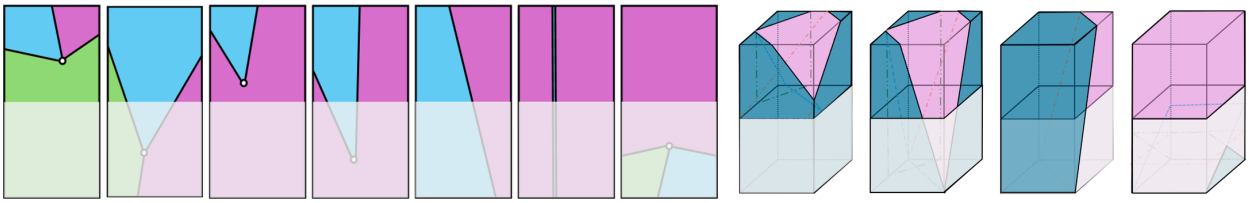


Fig. 1: Junction schematics reproduced from [7] (CC BY 4.0). Left: 2D junctions with M=3 wedge regions. Right: 3D junctions with M=2 regions formed by planes through a common vertex.

The full 3D FoJ objective couples all patches through a data-fidelity term and two inter-patch consistency regularizers,

$$\mathcal{L}(\theta) = \sum_p \Big[\underbrace{\|\hat{y}_p - y|_{\Omega_p}\|_2^2}_{\text{data}} + \lambda_B\, \mathcal{C}_p^B(\theta) + \lambda_C\, \mathcal{C}_p^C(\theta) \Big], \tag{2}$$

where $\mathcal{C}^B$ and $\mathcal{C}^C$ penalize disagreement between each patch's local boundary and intensity maps and the global estimates. Because every voxel belongs to multiple overlapping patches, the final denoised volume is obtained by averaging all patch renderings covering each voxel x:

$$\hat{x}(x) = \frac{1}{|\{p : x \in \Omega_p\}|} \sum_{p:\, x\in\Omega_p} \hat{y}_p(x). \tag{3}$$

Optimization proceeds in two stages: N_{init} *initialization* iterations of discrete coordinate-descent search over plane orientations and vertex positions, followed by N_{ref} gradient-based *refinement* iterations on Eq. (2).

3 Results and Discussion

We evaluate 3D FoJ across the three domains against identical classical baselines: 3D Gaussian smoothing (σ=1), a 3^3 median filter, and 3D total-variation (TV) denoising [1, 2] swept over weights $w \in \{0.05, 0.1, 0.2\}$. FoJ runs at its default schedule with patch sizes $R \in \{6, 8, 12\}$ and stride s=2. In addition to 3D PSNR, we measure a boundary F1-score on gradient-magnitude edge voxels to quantify interface preservation.

3.1 Reproducing the sparse-view CT ablation

Before evaluating new domains, we reproduce the sparse-view CT ablation of the original study (Table 3 of [7]), in which the 256^3 *engine* volume is reconstructed from 20 cone-beam projections corrupted by Poisson noise ("P50") and then denoised with 3D FoJ at $R \in \{4, 6, 8, 10, 12\}$. The released repository contains the FoJ model but not the CT simulation, view angles, or reconstruction

settings, so we rebuild the pipeline following the R^2-Gaussian benchmark [8] whose data generator the paper used: TIGRE [9] cone-beam geometry, 20 stratified-jitter view angles in $[0, 2\pi)$, and 60 CGLS iterations, followed by the released FoJ implementation (M=2, stride 2). Fig. 2 shows the three low-SNR applications demonstrated in the original study.

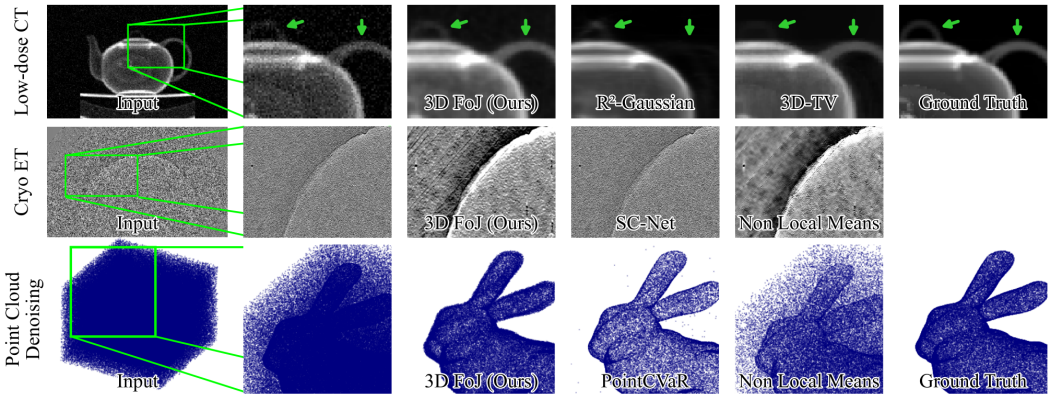


Fig. 2: Figure 1 of the original paper, reproduced from [7] (CC BY 4.0): 3D FoJ applied to low-dose CT (top row), cryo-ET (middle row), and point-cloud denoising (bottom row).

	R				
Metric	4	6	8	10	12
Reported PSNR	11.33	14.37	16.18	17.17	17.70
Ours PSNR	14.72	16.93	17.99	18.58	**18.86**
Δ PSNR	+3.39	+2.56	+1.81	+1.41	+1.16
Reported MS-SSIM	0.762	0.784	0.796	0.798	0.800
Ours MS-SSIM	0.786	0.829	0.854	0.870	**0.880**
Δ MS-SSIM	+0.024	+0.045	+0.058	+0.072	+0.080

Table 1: Reproduction of the engine-volume CT ablation of [7] (20 views, P50 Poisson noise). Ours uses the released M=2 implementation on the unclamped CGLS-60 output; Δ is ours minus reported.

Table 1 compares the reported denoising quality with our reproduction. The qualitative trend reproduces exactly: quality increases monotonically with patch size, R=12 is best, and the marginal gains shrink with each step (+2.2, +1.1, +0.6, +0.3 dB). Our absolute PSNR sits 1.2–3.4 dB above the reported values, consistent with the region count difference (M=2 in the released code versus M=3 in the paper) and reconstruction details the paper does not specify; the gap narrows with R as the wedge approximation becomes the dominant error. The released implementation thus behaves as documented on the published benchmark, and the domain study below uses it as-is.

3.2 Porous-media micro-CT (Bentheim sandstone)

We first test a 192^3 central crop of Bentheim sandstone micro-CT [10] (5.85 μm voxels, normalized to $[0, 1]$) under zero-mean additive Gaussian noise at $\sigma \in \{15\%, 30\%, 40\%, 55\%\}$ of the dynamic range. Because quartz grains and pore spaces form near-constant phases spanning 10–35 voxels separated by sharp interfaces, the volume satisfies both the unbiased-input and patch-scale piecewise-flat requirements of Eq. (1), isolating the effect of the noise regime.

σ	Noisy	Gaussian	Median	TV (best w)	FoJ (best R)
15%	16.89	**28.51**	27.01	27.65	26.55
30%	12.00	25.26	22.26	**26.03**	25.64
40%	10.28	23.39	20.04	23.32	**24.93**
55%	8.76	21.05	17.60	16.30	**24.29**

Table 2: 3D PSNR (dB) on Bentheim sandstone micro-CT. Classical filters lead at moderate noise ($\sigma \leq 30\%$), whereas FoJ overtakes all baselines once input PSNR drops below ∼11 dB ($\sigma \geq 40\%$).

Table 2 and Fig. 4 show the resulting crossover. At σ=15% and 30% (input PSNR 16.89 and 12.00 dB), local statistics remain reliable enough for Gaussian and TV filtering to recover the signal, and the coarse three-plane wedge approximation introduces more bias than the noise it removes. Once the input falls below ≈11 dB PSNR (σ=40% and 55%), classical filters either under-smooth the noise or blur across grain boundaries, whereas averaging regional means over R^3 wedges suppresses the variance while keeping planar interfaces intact. At σ=55%, FoJ exceeds the best classical baseline by +3.24 dB and retains a boundary-F1 of 0.85–0.88 compared with 0.54 for TV (Fig. 3). Within this compatible regime, the optimal patch size shifts from R=6 at $\sigma \leq 40\%$ to R=8 at σ=55%, before saturating at R=12 (23.75 dB) as patches begin to exceed individual grain geometry.

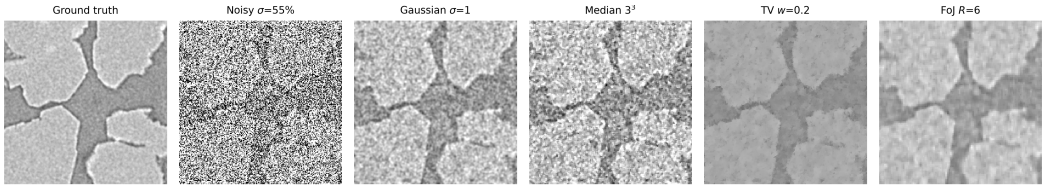


Fig. 3: Bentheim micro-CT mid-slice at σ=55% Gaussian noise. FoJ (R=6) preserves sharp grain–pore interfaces that Gaussian, median, and TV filtering blur.

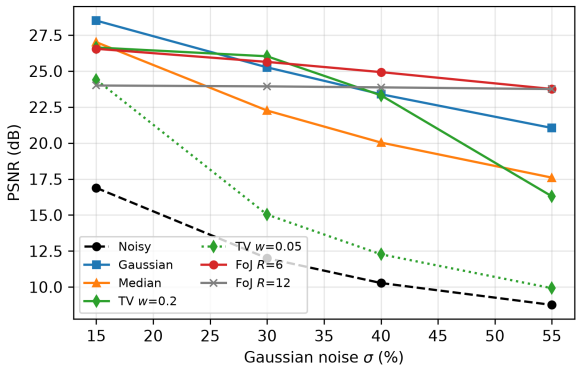


Fig. 4: PSNR versus noise level on Bentheim micro-CT. FoJ (red) overtakes the classical baselines between σ=30% and 40%.

3.3 Volumetric MRI phantom (BrainWeb T1)

To test what happens when structures are larger than the patch scale but violate piecewise flatness, we evaluate the BrainWeb T1 brain phantom [11] (181×217×181, scaled to $[0, 1]$) under Rician noise at $\sigma \in \{3\%, 5\%, 7\%, 9\%, 13\%\}$. Unlike porous rock, anatomical tissue exhibits smooth intensity gradients within white and gray matter rather than flat plateaus.

σ	Noisy	Gaussian	Median	TV (best w)	FoJ R=6	FoJ R=8	FoJ R=12
3%	29.19	28.69	**30.80**	29.85	25.67	24.02	21.56
5%	24.69	26.10	**27.25**	26.62	24.14	22.66	20.59
7%	21.74	23.80	**24.58**	24.00	22.52	21.43	19.78
9%	19.55	21.87	**22.48**	21.94	20.99	20.19	18.88
13%	16.34	18.77	**19.28**	18.68	18.31	17.86	17.05

Table 3: 3D PSNR (dB) on the BrainWeb T1 MRI phantom under Rician noise. Because smooth tissue gradients cannot be represented by constant wedges, FoJ trails classical filtering at every noise level and degrades monotonically as patch size R increases.

As Table 3 shows, replacing each wedge with its regional mean (Eq. (1)) quantizes smooth tissue variations into piecewise-constant steps. At σ=3%, FoJ (R=6) reduces PSNR below the noisy input itself (25.67 vs. 29.19 dB) and trails the 3^3 median filter by 5.13 dB. Although the gap narrows as noise grows, reaching 0.97 dB at σ=13% as variance reduction partially offsets model bias, FoJ never overtakes any classical baseline, and larger patches (R=8, 12) consistently perform worse because they force constant values over wider tissue regions. Using FoJ as a plug-and-play denoiser inside 4×-undersampled single-coil MRI reconstruction shows the same limitation, reaching 27.99 dB compared with 30.39 dB for TV-PnP.

3.4 Sub-patch binary porous media (sphere pack and Berea)

Finally, we test two binary porous-media volumes, a synthetic sphere pack and a segmented Berea sandstone scan [12], under additive Gaussian noise at $\sigma \in \{15\%, 30\%\}$. Both volumes are strictly piecewise-constant (binary solid and pore phases in $\{0, 1\}$), but their characteristic pore and grain features span only about 2 voxels, well below the smallest tested patch size (R=6).

Volume	σ	Noisy	Gaussian	Median	TV (best w)	FoJ R=6	FoJ R=8	FoJ R=12
Sphere pack	15%	19.49	13.90	17.63	**23.09**	12.10	10.67	8.90
Sphere pack	30%	13.48	13.77	15.52	**17.82**	12.04	10.65	8.89
Berea	15%	**19.49**	7.49	5.88	17.73	7.10	6.91	6.72
Berea	30%	13.48	7.45	6.37	**13.61**	7.09	6.91	6.72

Table 4: 3D PSNR (dB) on binary porous-media volumes with ∼2-voxel feature scale. Because sub-patch features cannot be partitioned by three planes, each wedge mean collapses toward the global mean.

Table 4 and Fig. 5 show that when the underlying geometry is finer than the patch support Ω_p, the wedge parameterization fails completely. Three planes through a single vertex cannot separate dozens of 2-voxel pores inside an R^3 patch, so every wedge covers a mixture of solid and pore voxels and Eq. (1) assigns each region an intensity near the patch mean. Averaging overlapping patches in Eq. (3) then produces a nearly flat volume: on Berea at σ=15%, the output standard deviation drops from 0.49 to 0.09 and PSNR collapses from the noisy input's 19.49 dB to 7.10 dB (R=6) and 6.72 dB (R=12).

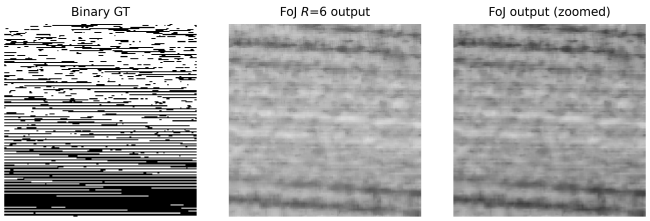


Fig. 5: Collapse on sub-patch-scale binary structure (Berea, σ=15%). Features of ∼2 voxels cannot be represented inside R=6 patches, so regional means collapse the output toward the volume mean.

4 Conclusion

Evaluating 3D Field of Junctions across three volumetric domains outside its original benchmark shows that its effectiveness is governed directly by the regional-mean wedge formulation in Eqs. (1)–(3). Because each patch is modeled as constant-valued wedges whose intensities are local voxel means, any systematic offset in the input is preserved rather than removed, and only structures that are piecewise-flat at or above the patch scale R can be represented. When the volume satisfies these structural conditions, as in grayscale Bentheim sandstone micro-CT where grains and pores span tens of voxels with sharp interfaces, 3D FoJ overtakes Gaussian, median, and TV filtering once the input PSNR falls below $\approx$11 dB, achieving a +3.2 dB advantage and superior boundary retention at σ=55% because the variance removed across each wedge outweighs the model's approximation bias. Conversely, when the tissue is smooth (BrainWeb MRI), constant wedges quantize real intensity gradients and underperform classical filters at every tested noise level; when the features are smaller than the patch (binary sphere pack and Berea sandstone), wedges cannot resolve individual pores and the output collapses toward the global mean. 3D FoJ is therefore a specialized, training-free prior for low-SNR volumes whose phase structure is piecewise-flat and larger than the patch scale.

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